ALBERT-LUDWIGS-UNIVERSITÄT FREIBURG

ON THE EFFICIENCY OF EXCITONIC ENERGY TRANSPORT

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Introduction

experimental evidence[1,2] for quantum coherence in the Fenna-Matthews-Olson (FMO) protein of green sulfur bacteria

energy shuttle

photon ⟶ antenna ⟶ FMO ⟶ reaction center

efficiency 95% of incoming energy reaches the reaction center

transport mechanism incentive to employ quantum coherence?

energy transport[3] mediated by 7 (or 8) molecular sites with discrete electronic energy levels

structural stiffness provided by surrounding protein scaffold

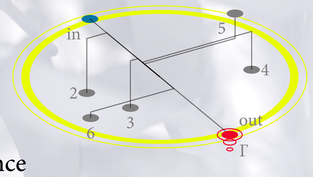
coherence preserved by nontrivial, noisy environment

Candidate Mechanisms—The Role of Quantum Coherence

assistance of dephasing noise[4,5]	vs	enhancement by quantum interference[8,9]
ambient dephasing noise suppresses detrimental destructive interference, *i.e.* quantum localization	**PREMISE**	interference sensitive to system parameters, fluctuations hallmark of coherence
optimize bath to foster energy transport through disordered system	**OBJECTIVE**	*optimize system* such that constructive interference enables efficient transport
find optimal local dephasing rates $\{\gamma_i\}$ for Lindbladian $L_{\text{deph}}(\rho) = 4\sum_i \gamma_i \left(\sigma_{+,i}\,\sigma_{-,i}\,\rho\,\sigma_{+,i}\,\sigma_{-,i} - \frac{1}{2}\{\sigma_{+,i}\,\sigma_{-,i}, \rho\}\right)$	**CHALLENGE**	find optimal spatial configuration with $\mathfrak{T}_{\text{opt}} \ll \widetilde{\mathfrak{T}}$
most configurations per se inefficient, can be assisted by noise at the expense of coherence	**DIAGNOSIS**	optimal coherent transport of rare configurations *most efficient*[6,7]

Ensembles of Dipolar Networks

Spatial Disorder



single excitation transport from fixed node 'in' to fixed node 'out'

uniform distribution of intermediate nodes in d-ball, minimum distance $r_{\min} \leq r_{i,j} \leq r_{\text{in,out}}$

Hamiltonian $H = \sum_i E_i\,\sigma_{+,i}\,\sigma_{-,i} + \sum_{i\neq j} V_{i,j}\,\sigma_{+,i}\,\sigma_{-,j}$ with disordered, correlated $V_{i,j} = \mathfrak{A}\left(\boldsymbol{d}_i, \boldsymbol{d}_j, \vartheta_{i,j}, \varphi_{i,j}\right) r_{i,j}^{-3}$

reference time $T \equiv$ half a Rabi oscillation between $|\text{in}\rangle$ and $|\text{out}\rangle$

Transport Efficiency

irreversible transfer to sink at 'out' with rate Γ described by $L_{\text{sink}}(\rho) = \Gamma\left(\sigma_{-,\text{out}}\,\rho\,\sigma_{+,\text{out}} - \frac{1}{2}\{\sigma_{+,\text{out}}\,\sigma_{-,\text{out}}, \rho\}\right)$, accounts for processing of excitation at reaction center

sink drain $\mathfrak{p}(t) = \Gamma\int_0^t \langle\text{out}|\rho(t')|\text{out}\rangle\,\mathrm{d}t' \xrightarrow{t\to\infty} 1$

efficiency measure average transfer time $\mathfrak{T} = \int_0^\infty t\,\dot{\mathfrak{p}}(t)\,\mathrm{d}t$

heavy-tailed probability density $f_{\mathfrak{T}}(\mathfrak{t}) \propto \mathfrak{t}^{-\mu}$, $\mu > 1$, $\mathfrak{t} \to \infty$, $\mathfrak{T}$ can be arbitrarily long due to destructive interference

typical value given by median $\widetilde{\mathfrak{T}}$ defined by $\int_0^{\widetilde{\mathfrak{T}}/T} f_{\mathfrak{T}}(\mathfrak{t})\,\mathrm{d}\mathfrak{t} = 1/2$

Topography of Transport And Dephasing

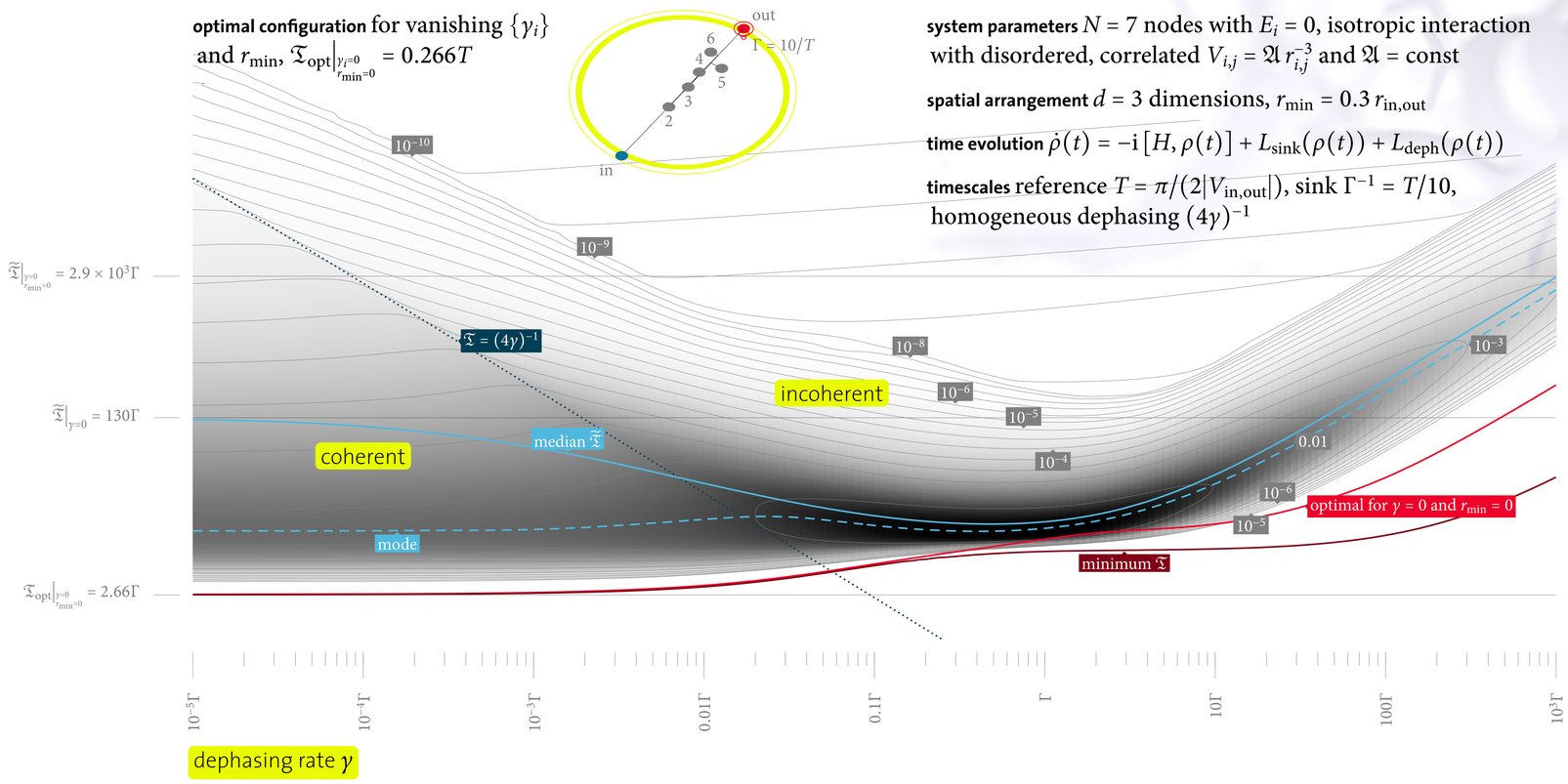
optimal configuration for vanishing $\{\gamma_i\}$ and $r_{\min}$, $\mathfrak{T}_{\text{opt}}\big|_{\substack{\gamma_i=0\\ r_{\min}=0}} = 0.266T$

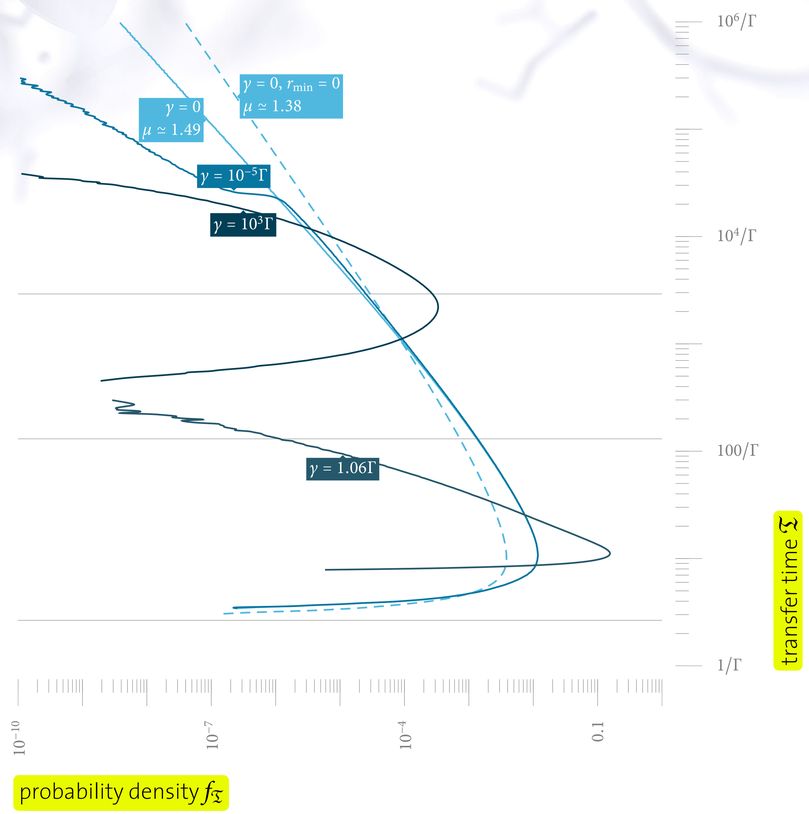
system parameters $N = 7$ nodes with $E_i = 0$, isotropic interaction with disordered, correlated $V_{i,j} = \mathfrak{A}\,r_{i,j}^{-3}$ and $\mathfrak{A} = \text{const}$

spatial arrangement $d = 3$ dimensions, $r_{\min} = 0.3\,r_{\text{in,out}}$

time evolution $\dot{\rho}(t) = -\mathrm{i}\,[H, \rho(t)] + L_{\text{sink}}(\rho(t)) + L_{\text{deph}}(\rho(t))$

timescales reference $T = \pi/(2|V_{\text{in,out}}|)$, sink $\Gamma^{-1} = T/10$, homogeneous dephasing $(4\gamma)^{-1}$





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